first steps with quarto

prof. dr. nicolas meseth

1. one source
2. writing
3. code and results
4. the scholarly apparatus
5. rendering

every time the data change, someone copies the chart into the report again, unless the report computes the chart itself.



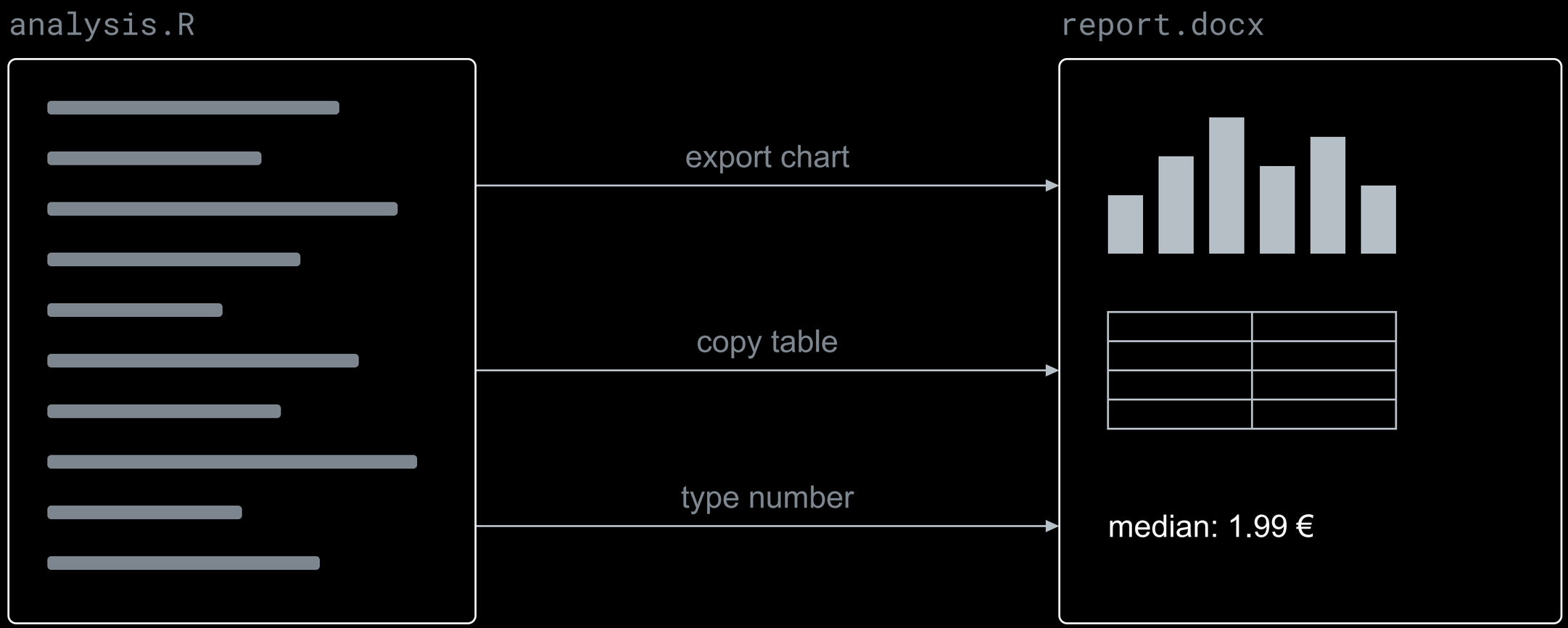
image: ai-generated (gpt-image-2.5-flare)

part 1

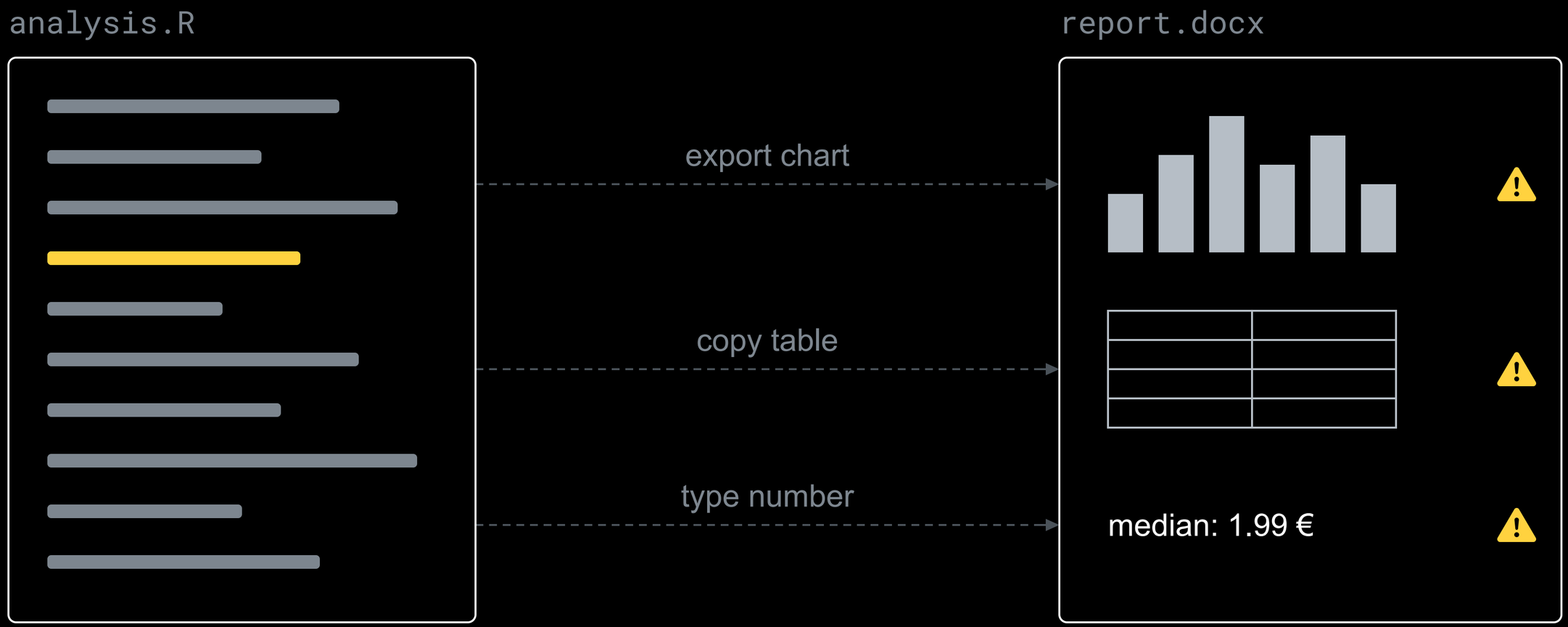
one source

text, code and results in one file

copy and paste



copy and paste



what do you write longer texts in today?

word

google docs

latex or typst

markdown

something else



live.hsos.uber.space/q

what do you write longer texts in today?

- word
- google docs
- latex or typst
- markdown
- something else

live.hsos.uber.space/q

````
---
title: "What is milk worth?"
format: typst
---

## Introduction
Milk is one of the few foods
whose price many know by heart.

```{r}
survey |>
  count(diet)
```
````

milk-prices.qmd

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metadata

about the whole document

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markdown
the text, with a few marks

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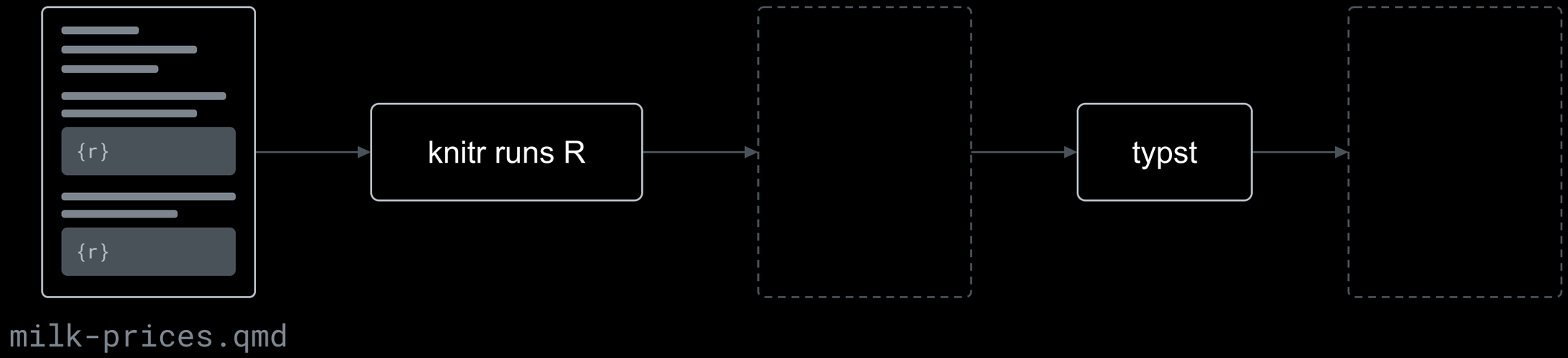
markdown

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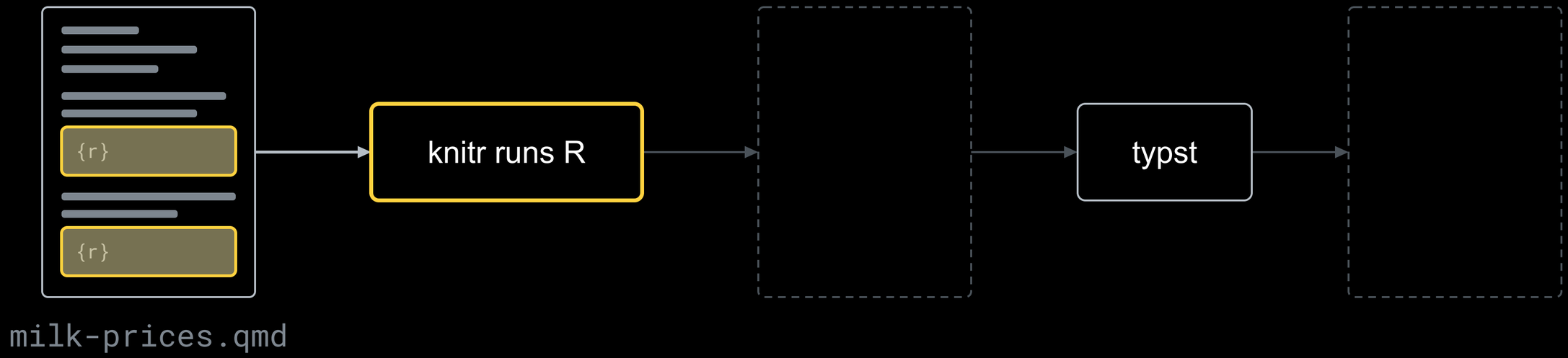
R code

runs when the document renders

what rendering does



what rendering does



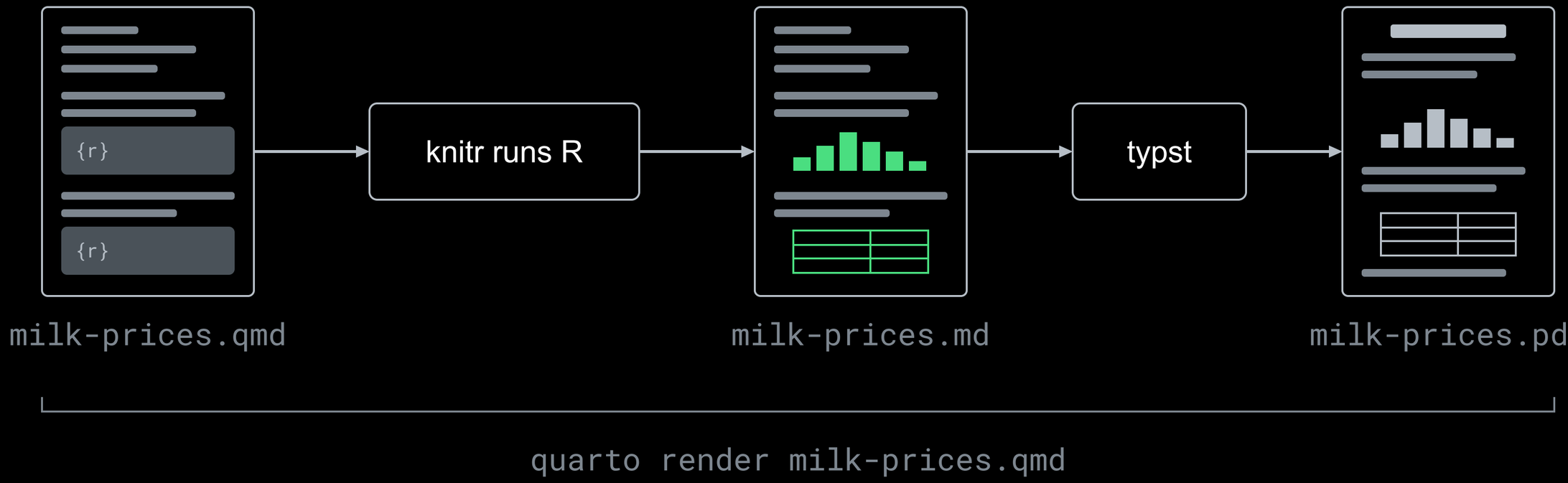
what rendering does



what rendering does



what rendering does



the document is the analysis, so a number in the text can only be wrong if the code is wrong.

part 2

writing

header, markdown, references, math

```yaml
---
title: "What is milk worth?"
author: "Group 4"
date: today                                  # the day of rendering
lang: en                                     # english hyphenation and labels
abstract: |
  We ask from which price on a litre
  of milk is too expensive for shoppers.
format:
  typst:                                     # the PDF is set by typst
    papersize: a4                            # without it, the page is US letter
    number-sections: true                    # numbers for the headings
execute:
  echo: false                                # results, but no code
bibliography: references.bib                 # the file with the sources
csl: apa                                     # the citation style
---
```

typed in the .qmd

```
## Method

This is **important**.

We ran a *survey*.

- one item
- another item

See [quarto.org](https://quarto.org).

A claim.^[a footnote]
```

in the PDF

1 Method

This is **important**.

We ran a *survey*.

- one item
- another item

See quarto.org.

A claim.[1]

[1] a footnote

sections you can point to

a heading with a label

```
## Method {#sec-method}
```

a reference to it in the text

```
As described in @sec-method, the respondents form a quota sample.
```

in the PDF

> As described in Section 2, the respondents form a quota sample and not a random one, so the thresholds describe the people who took part and not all of Germany.

math

typed in the .qmd

```
$$
\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i
$$
```

in the PDF

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

one dollar sign for math inside a sentence, two for a formula on a line of its own

part 3

code and results

chunks, figures, tables, numbers

a code chunk

````
```{r}
#| label: setup
#| message: false
library(tidyverse)

survey <- read_csv("milk_survey_compact.csv")
```
````

lines that start with #| are options for quarto, and r ignores them as comments

chunk options

option	code shown	runs	result shown
`echo: false`	no	yes	yes
`include: false`	no	yes	no
`eval: false`	yes	no	no
`warning: false`	yes	yes	yes, without warnings
`message: false`	yes	yes	yes, without messages

for the whole paper, set `echo: false` once in the header, under `execute:`

a figure

typed in the .qmd

````
```{r}
#| label: fig-threshold
#| fig-cap: "Price at which milk is too expensive."
survey |>
  filter(price_too_expensive <= 5) |>
  ggplot() +
  aes(x = price_too_expensive) +
  geom_histogram(binwidth = 0.25)
```
````

in the PDF

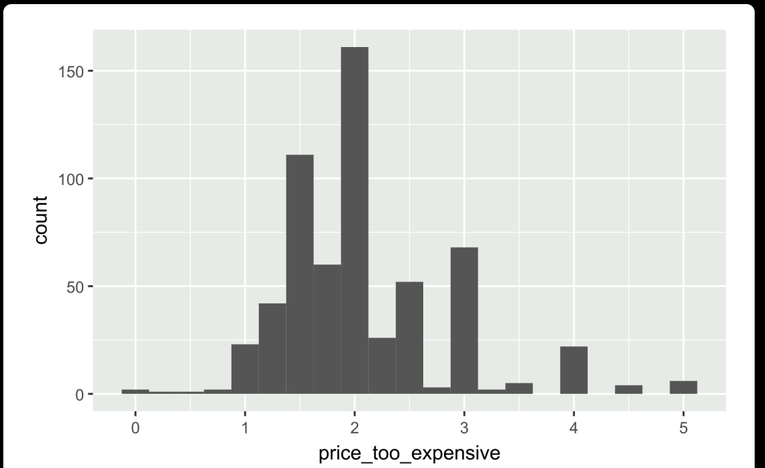


Figure 1: Price at which milk is too expensive.

a table

typed in the .qmd

````
```{r}
#| label: tbl-diet
#| tbl-cap: "Respondents by diet."
survey |>
  count(diet, sort = TRUE) |>
  knitr::kable()
```
````

in the PDF

Table 1: Respondents by diet.

diet	n
omnivore	1826
flexitarian	749
vegetarian	134
pescetarian	36
vegan	36
NA	28

numbers in the text

typed in the .qmd

```
The median threshold is
`{r} median(survey$price_too_expensive, na.rm = TRUE)`
euros.
```

numbers in the text

typed in the .qmd

```
The median threshold is
`{r} median(survey$price_too_expensive, na.rm = TRUE)`
euros.
```

in the PDF

The median threshold is 1.99 euros.

part 4

the scholarly apparatus

cross-references and citations

typed in the .qmd

```
@fig-threshold shows the answers up to five euros.

Most respondents eat meat, see @tbl-diet.
```

cross-references

typed in the .qmd

```
@fig-threshold shows the answers up to five euros.

Most respondents eat meat, see @tbl-diet.
```

in the PDF

Figure 1 shows the answers up to five euros.

Most respondents eat meat, see Table 1.

references.bib

```bibtex
@book{wickham2023,
  author = {Wickham, Hadley and
    Çetinkaya-Rundel, Mine and
    Grolemund, Garrett},
  title = {R for Data Science},
  edition = {2},
  publisher = {O'Reilly},
  year = {2023},
  url = {https://r4ds.hadley.nz}
}
```

references.bib

```
@book{wickham2023,
  author = {Wickham, Hadley and
    Çetinkaya-Rundel, Mine and
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  edition = {2},
  publisher = {O'Reilly},
  year = {2023},
  url = {https://r4ds.hadley.nz}
}
```

milk-prices.qmd

```
we summarise it [@wickham2023, chap. 10].
```

references.bib

```
@book{wickham2023,
  author = {Wickham, Hadley and
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}
```

milk-prices.qmd

```
we summarise it [@wickham2023, chap. 10].
```

milk-prices.pdf

we summarise it (Wickham et al., 2023, chap. 10).

Bibliography

Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for Data Science* (2nd ed.). O'Reilly. https://r4ds.hadley.nz/

which label lets you write @fig-prices in the text?

label: prices

label: fig-prices

label: figure-prices

fig-cap: prices

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which label lets you write @fig-prices in the text?

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live.hsos.uber.space/q

part 5

rendering

from the .qmd to the pdf

render

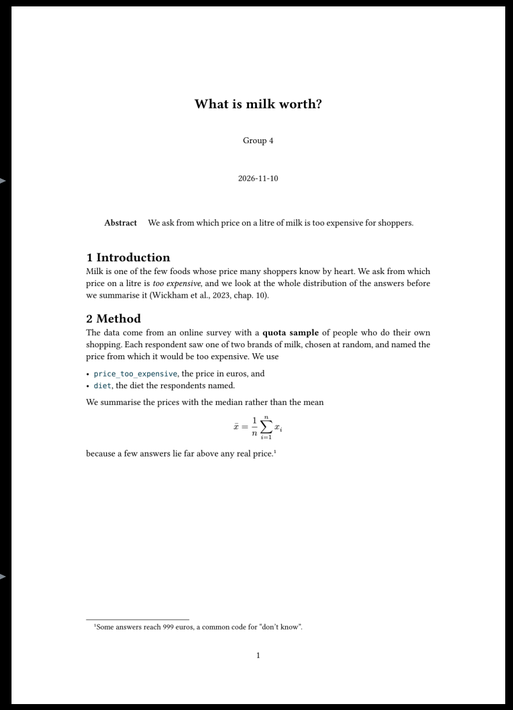
in positron

milk-prices.qmd

preview

in the terminal

```
quarto render milk-prices.qmd
```



What is milk worth?

Group 4

2026-11-10

Abstract We ask from which price on a litre of milk is too expensive for shoppers.

1 Introduction

Milk is one of the few foods whose price many shoppers know by heart. We ask from which price on a litre is *too expensive*, and we look at the whole distribution of the answers before we summarise it (Wickham et al., 2023, chap. 10).

2 Method

The data come from an online survey with a **quota sample** of people who do their own shopping. Each respondent saw one of two brands of milk, chosen at random, and named the price from which it would be too expensive. We use

- `price_too_expensive`, the price in euros, and
- `diet`, the diet the respondents named.

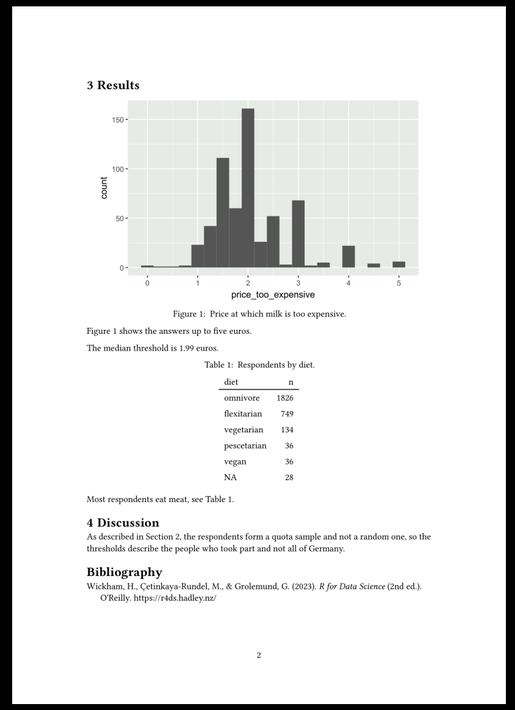
We summarise the prices with the median rather than the mean

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

because a few answers lie far above any real price.[1]

[1] Some answers reach 999 euros, a common code for "don't know".

1



3 Results

Figure 1: Price at which milk is too expensive.

Figure 1 shows the answers up to five euros.

The median threshold is 1.99 euros.

Table 1: Respondents by diet.

diet	n
omnivore	1826
flexitarian	749
vegetarian	134
pescetarian	36
vegan	36
NA	28

Most respondents eat meat, see Table 1.

4 Discussion

As described in Section 2, the respondents form a quota sample and not a random one, so the thresholds describe the people who took part and not all of Germany.

Bibliography

Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for Data Science* (2nd ed.). O'Reilly. https://r4ds.hadley.nz/

2

milk-prices.pdf

a package is not loaded

```
could not find function "ggplot"
```

when rendering fails

a package is not loaded

```
could not find function "ggplot"
```

a file is not found

```
does not exist in current working directory
```

when rendering fails

a package is not loaded	`could not find function "ggplot"`
a file is not found	`does not exist in current working directory`
the header is indented wrongly	`Validation of YAML front matter failed.`

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a label without fig- or tbl-	`?@fig-threshold`

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a label without fig- or tbl-	`?@fig-threshold`

---

no blank line before a list	the list becomes running text
no `papersize: a4`	the page is US letter

write the analysis once, render as often as you like, and every number in the paper stays the one the data give.



image: ai-generated (gpt-image-2.5-flare)